PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

THE GRATE BED POST

下篦床岗位

MTI-CHP-CPM-SOP-021_Rev. 000

Information Security Classification
B – Business Information

Rev.	Revision Status	Date	Prepared by	Reviewed by	Approved by
0	Issued for Use	24-APR-2024	*Approval as attached*	*Approval as attached*	*Approval as attached*
C0	Issued for Approval	24-APR-2024	Song Yuhua	Li Jianjun	Clark Wu
B0	Issued for Review	24-APR-2024	Song Yuhua	Li Jianjun	
A0	Issued for Internal Discipline Review	24-APR-2024	Song Yuhua		

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DOCUMENT CHANGE STATUS

No.	Page	Section	Change Log	Revision by	Revision date	Revision version
1.						
2.						
3.						

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the grate bed post under complete closure.

此程序涵盖操作员对完全关闭下的下篦床岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the grate bed post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the grate bed post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对下篦床岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定下篦床岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP 岗位操作规程
- Belt Conveyors (4303-BCV-001/005/006). 胶带机 (4303-BCV-001/005/006)
- High Efficiency Multi Tube Dust Collector (4303-DUC-001). 高效多管除尘器 (4303-DUC-001)
- Scrubbing Tower (4303-TOW-001-1). 一级空塔洗涤塔 (4303-TOW-001-1)
- Scrubbing Tower (4303-TOW-001-2). 二级填料洗涤塔 (4303-TOW-001-2)
- Circulating Water Pump (4303-PSO-001/002/003/004). 循环泵 (4303-PSO-001/002/003/004)
- Pressure Filter (4303-FIL-001). 压滤机 (4303-FIL-001)
- Pressure Filter Pump (4303-PSL-001/002). 压滤泵 (4303-PSL-001/002)
- Screwed Conveyor (4303-CNV-001). 螺旋输送机 (4303-CNV-001)
- Exhaust Fan (4303-FAN-005/201). 引风机 (4303-FAN-005/201)
- Bag Filter (4303-DUC-201). 布袋除尘器 (4303-DUC-201)
- Pipe Chain Conveyor (4303-PCC-001). 管链输送机 (4303-PCC-001)

4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服

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防尘口罩

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Dust; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Object falling from high altitudes; 7. Noise. 1. 机械伤害； 2. 灰尘； 3. 触电； 4. 湿滑的条件，绊倒危险； 5. 窒息； 6. 高空坠物； 7. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear dust mask as required; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear labor protection goods and check the safety of stacking items on the operation platform; 7. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 按要求佩戴防尘口罩； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 穿戴好劳保护品，检查各平台的物品堆放安全性； 7. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4303-ADG-001 ~ 028	Double Layer Ash Valve双层卸灰阀
4303-RAP-001	Rap Device振打器
	Star Discharger星型卸料阀
4303-PSU-001	Submerged Pump液下泵
	Clean the site tools清扫现场工具

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7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Belt Conveyors (4303-BCV-001/005/006) 检查胶带机(4303-BCV-001/005/006)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 2. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好; 3. 全面检查胶带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好; 4. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏; 5. 检查承受冲击载荷部位的缓冲托辊是否完好; 6. 检查胶带机清扫器安装是否正常;

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№	Stage	Key Points
		7. 装设的电器设施及信号装置是否完好和反应灵敏； 8. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 9. 检查无误并与前后岗位联系后，按下启动按钮； 10. 运转过程中，检查胶带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
3	Inspect High Efficiency Multi Tube Dust Collector (4303-DUC-001) 检查高效多管除尘器(4303-DUC-001)	1. Check whether the connection part is firm; 2. Check the air tightness of the dust collector, the flue, ash pipe and the ash valve; 3. Check all flanges and tie them with bolts; 4. Check whether the high and low material level detector of the dust collector is normal; 5. Check to ensure that the outlet pressure and temperature are normal. 2. 检查各连接部位是否牢固； 3. 检查除尘器与进出烟道、灰斗等结合部位及排灰阀的气密性； 4. 检查所有法兰和法兰连接时中间需夹石棉绳，并用螺栓拴紧； 5. 检查除尘器灰斗高、低料位检测仪是否正常； 6. 检查确保出口压力、温度正常。
4	Inspect Scrubbing Tower (4303-TOW-001-1) 检查一级空塔洗涤塔 (4303-TOW-001-1)	1. Remove all sundries in the first tower washing tower and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the new water stop valve (25GA) before the circulation tank of the first tower, inject the first tower into the new water, open the circulation pump (4303-PSO-001/002) butterfly valve (350BF, 300BF), and open the circulation pump (4303-PSO-001/002); 5. Observe the spray washing situation of the first empty tower, whether the spray amount and density meet the requirements; 6. Check whether the spray support ring, fog gate plate, screen separator and nozzle assembly are normal, and the spray remover and nozzle are not blocked; 7. Check whether the packing, compactness and height of the packing meet the requirements, and whether the spray dispersion is uniform; 8. Check whether the circulation liquid inlet and outlet, access outlet, sewage outlet, gas inlet and outlet, pump outlet, water supply outlet, manhole are normal, whether the connection bolts are fastened, whether liquid leakage or air leakage; check whether the tower body is leakage or air leakage;

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№	Stage	Key Points
		9. Empty the inspection water in the washing tower of the first-stage empty tower. 2. 清除一级空塔洗涤塔等设备内的一切杂物； 3. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常； 4. 检查各液体循环管线、气体管路、阀门是否完好； 5. 打开一级空塔洗涤塔循环槽前新水截止阀(25GA)，将一级空塔洗涤塔注入新水，打开循环泵(4303-PSO-001/002)出入口蝶阀(350BF,300BF)，开启循环泵(4303-PSO-001/002)； 6. 观察一级空塔洗涤塔喷淋洗涤情况，喷淋量、喷淋密度是否符合要求； 7. 检查喷淋支座环、除雾栅板、丝网除沫器、喷嘴组件是否正常，除沫器和喷嘴有无堵塞现象； 8. 检查填料排布、密实度和高度是否符合要求，喷淋液散布是否均匀； 9. 检查循环液进出口、检修口、排污口、气体进出口、泵口、补水口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 10. 放空一级空塔洗涤塔内的检查用水。
5	Inspect Scrubbing Tower (4303-TOW-001-2) 检查二级填料洗涤塔 (4303-TOW-001-2)	1. Remove all sundries in the secondary packing washing tower and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the new water stop valve (25GA) before the secondary filler washer circulation tank, inject the secondary filler washer into the new water, open the circulation pump (4303-PSO-003/004) inlet butterfly valve (350BF, 300BF), and open the circulation pump (4303-PSO-003/004); 5. Observe the spray washing situation of the secondary filler washing tower, and whether the spray amount and spray density meet the requirements; 6. Check whether the spray support ring, fog gate plate, screen separator and nozzle assembly are normal, and the spray remover and nozzle are not blocked; 7. Check whether the packing, compactness and height of the packing meet the requirements, and whether the spray dispersion is uniform; 8. Check whether the circulating fluid inlet, access outlet, sewage outlet, gas outlet, pump outlet, water supply outlet, manhole are normal, whether the connection bolts are fastened, whether leakage or leakage; check whether the tower body leakage or leakage; 9. Empty the inspection water in the secondary packing washer. 1. 清除二级填料洗涤塔等设备内的一切杂物； 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常；

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№	Stage	Key Points
		3. 检查各液体管线、气体管路、阀门是否完好； 4. 打开二级填料洗涤塔循环槽前新水截止阀(25GA)，将二级填料洗涤塔注入新水，打开循环泵(4303-PSO-003/004)出入口蝶阀(350BF,300BF)，开启循环泵(4303-PSO-003/004)； 5. 观察二级填料洗涤塔喷淋洗涤情况，喷淋量、喷淋密度是否符合要求； 6. 检查喷淋支座环、除雾栅板、丝网除沫器、喷嘴组件是否正常，除沫器和喷嘴有无堵塞现象； 7. 检查填料排布、密实度和高度是否符合要求，喷淋液散布是否均匀； 8. 检查循环液进口、检修口、排污口、气体进出口、泵口、补水口、人孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 9. 放空二级填料洗涤塔内的检查用水。
6	Inspect Circulating Water Pump (4303-PSO-001/002/003/004) 检查循环泵(4303-PSO-001/002/003/004)	1. Check whether the installation of pump inlet pipelines and auxiliary pipelines, flanges and valves meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft in 3 circles to be flexible without blocking; 3. Check whether the lubricating oil level is normal, and the oil level should be between 1/2 and 2/3 of the oil mirror; 4. Check whether the seal of the pump is normal; 5. Point and move to confirm the correct steering; 6. Check whether the pump control cabinet voltmeter, signal lights and other instrument indicators are normal; 7. Check whether there is air in the pump unit, if it should be excluded; 8. Check whether the pump head is under the liquid level of the first-stage empty tower (4303-TOW-001-1) and the second-stage filling tower (4303-TOW-001-2); 9. Check that the conveying pump is in normal state, and the outlet butterfly valve (350BF) is temporarily closed after flexible flexibly; 10. Check the backup pump outlet butterfly valve (350BF) completely. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好； 2. 手转动泵轴3圈，应灵活无阻滞； 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3 之间； 4. 检查泵的密封水状况是否正常； 5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 检查泵头是否处于一级空塔洗涤塔(4303-TOW-001-1)和二

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№	Stage	Key Points
		级填料洗涤塔(4303-TOW-001-2)液位下； 9. 检查输送泵处于正常状态，出口蝶阀(350BF)开启灵活后暂时处于关闭状态； 10. 检查备用泵出口蝶阀(350BF)关闭状态。
7	Inspect Pressure Filter (4303-FIL-001) 检查压滤机(4303-FIL-001)	1. Check whether the number of filter plates is sufficient, whether there is damage, whether the filter plate is clean, and whether the placement meets the requirements; 2. Check whether the filter cloth is folded, whether it is damaged, and whether the filter performance is good; 3. Check whether the pressure gauge of the mine pipeline is normal; 4. Check whether the connection is close and whether there is leakage; 5. Check whether the oil of the filter press is sufficient and whether the oil level meets the requirements (1 / 2~1 / 3); 6. Check whether the hydraulic oil top device is normal; 7. Check whether the emergency stop device is normal; 8. Check whether the pressure gauge of the hydraulic oil is normal and whether the pointer is in the specified interval; 9. Check whether the import and export of the filter press are smooth; 10. Automatic plate drawing device; 11. Whether the water connection plate is flexible and easy to use. 1. 检查滤板数量是否足够，有无破损，滤板是否清洁，安放是否符合要求； 2. 检查滤布是否折叠，有无破损，过滤性能是否良好； 3. 检查给矿管路压力表是否正常； 4. 检查各处联接是否紧密，有无泄露； 5. 检查压滤机油是否足够，油位是否符合要求(1/2 ~ 1/3)； 6. 检查液压油顶装置是否正常； 7. 检查急停装置是否正常； 8. 检查液压油的压力表是否正常，指针是否在规定的区间； 9. 检查压滤机各进出口是否畅通； 10. 检查自动拉板装置良好； 11. 接水盘是否灵活好用。
8	Inspect Pressure Filter Pump (4303-PSL-001/002) 检查压滤泵(4303-PSL-001/002)	1. Check whether the installation of pump inlet and auxiliary pipelines, flanges and valves meet the requirements, whether the anchor bolts and ground wires are good, and whether the coupling is installed; 2. Turn the pump shaft for 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal, the oil level is between 1/2 and 2/3 of the oil mirror; 4. Check whether the sealing condition of the pump is normal; 5. Click and move to confirm the correct steering; 6. Check whether the pump control cabinet voltmeter, signal

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		lights and other instrument indicators are normal; 7. Check whether there is air in the pump unit, if it should be excluded; 8. Check whether the pump head is under the liquid level of the first-stage empty tower washing tower (4303-TOW-001-1) and the second-stage packing washing tower (4303-TOW-001-2); 9. Check that the conveying pump is in a normal state, and the outlet clip pipe valve (100PN) and check valve (100 CH) are temporarily closed after opening flexibly; 10. Check the backup pump outlet check valve (100 CH) completely. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器是否装好； 2. 手转动泵轴3圈，应灵活无阻滞； 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3 之间； 4. 检查泵的密封水状况是否正常； 5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 检查泵头是否处于一级空塔洗涤塔(4303-TOW-001-1)和二级填料洗涤塔(4303-TOW-001-2)液位下； 9. 检查输送泵处于正常状态，出口夹管阀(100PN)和止回阀(100CH)开启灵活后暂时处于关闭状态； 10. 检查备用泵出口止回阀(100CH)全关。
9	Inspect Screwed Conveyor (4303-CNV-001) 检查螺旋输送机(4303-CNV-001)	1. Check whether the lubricating grease in the bearing is appropriate; 2. Check whether the oil quality and quantity of the lubricating oil in the reducer are normal; 3. Check whether the bolts are fastened; 4. Check whether the cover plate of the screw conveyor must be tightly covered and whether the conveyor is clean; 5. Check whether the spiral blade is deformed ; 6. Determine whether the rotation direction of the motor meets the work requirements. 1. 检查轴承内润滑脂是否合适； 2. 检查减速机内的润滑油油质、油量是否正常； 3. 检查各螺栓是否紧固； 4. 检查螺旋输送机的盖板必须盖严，输送机内是否清理干净； 5. 检查螺旋叶片是否变形； 6. 判明电机旋转方向是否符合工作要求。
10	Inspect Exhaust Fan (4303-FAN-005/201) 检查引风机(4303-	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The fan air outlet valve is normal, and the fan inlet valve is closed;

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№	Stage	Key Points
	FAN-005/201)	3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 1. 手动盘车2～3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 风机出风阀门正常，风机进口阀门关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常； 5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)。
11	Inspect Bag Filter (4303-DUC-201).检查布袋除尘器 (4303-DUC-201)	1. Whether the electrical high and low voltage system and PLC control are normal; 2. Whether the inlet valve of the filter chamber of each unit is open; 3. Whether all the solenoid pulse valves are normal and the action sequence is correct; 4. All measurement circuits are normal and signal; 5. All control valves are in the automatic mode and in the desired position; 6. Open all inspection entrances and doors to check whether the equipment moving parts are stuck in dust or external objects; after inspection, close all inspection population and doors and check all pipes and equipment inspection holes for closure; 7. Whether the sealing strip of the air chamber is sealed, and whether the fastening bolt of air box sealing cover is in good condition; 8. Connect the pressure measuring device to the pressure measuring port of the bag filter; 9. Check whether the bag back blowing pressure is 0.4MPa~0.5MPa; 10. Check whether the water drain trap is normal; 11. Check the operation condition of the ash removal device of the bag filter. 1. 电气高低压系统、PLC控制是否正常； 2. 除尘器各单元过滤室的进口阀门是否都打开； 3. 所有电磁脉冲阀是否正常，动作顺序正确； 4. 所有测量回路是否正常、有信号；

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		5. 所有提升阀处于自动模式和需要的位置； 6. 打开所有检查入口、门，检查设备运动部件是否被灰尘或外来物卡住；检查后，关闭所有检查人口、门，并检查所有管道和设备检查孔是否关闭； 7. 反吹风气室密封条是否是密封完好，气箱密封盖紧固螺栓是否完好； 8. 接通测压装置到布袋除尘器的测压口； 9. 检查布袋反吹风压力是否是0.4MPa ~ 0.5MPa； 10. 检查疏水阀是否正常； 11. 检查布袋除尘器排灰装置工作情况。
12	Inspect Pipe Chain Conveyor(4303-PCC-001) 检查管链输送机(4303-PCC-001)	1. Check whether the bolt connection of the pipe wall and the chain plate is loose; 2. Check whether the tightness of the traction chain is normal; 3. Check whether the chain plate conveyor should inject lubricant in time at each lubrication point is normal and whether oil leakage; 4. Check whether the transmission parts are falling off, are not firmly tightened, or have abnormal friction; 5. Determine whether the rotation direction of the motor meets the work requirements; 6. Check whether the conveying roll is partial. 1. 检查管壁、链板的螺栓连接是否有松动现象； 2. 检查牵引链的松紧度是否正常； 3. 检查链板输送机在各润滑点应及时注入润滑剂是否正常，是否漏油； 4. 检查传动部件是否有脱落、拧紧不牢或摩擦异常； 5. 判明电机旋转方向是否符合工作要求； 6. 检查输送时卷料是否偏料。
13	Check the other Equipment and Facilities 检查其它设备设施	1. Double Layer Ash Valve(4303-ADG-001 ~ 028) a. Turn on the power supply and check whether the motor steering direction is consistent with the arrow mark; b. Check whether the electric double-layer ash discharge valve weight block stops and returns, whether the swing rod is loose and falls off, and whether the flange connection bolt is loose and falls off; c. Fill the bearing base with lubricating grease (general lithium grease, 30g /). 1. 双层卸灰阀(4303-ADG-001 ~ 028) a. 接通电源，检查电机转向与箭头标示方向是否一致； b. 检查电动双层卸灰阀配重块止退销、摆杆是否松动脱落，法兰连接螺栓是否有松动脱落； c. 轴承座加注润滑油脂(通用锂基脂,30g/处)。 2. Rap Device(4303-RAP-001) a. Check whether the connection of each part is fastened; b. Check whether the rotation direction is consistent with

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		the mark direction. 2. 振打器(4303-RAP-001) a. 检查各部位连接是否紧固； b. 检查旋转方向是否与标记方向一致。 3. Star Discharger a. Check that the rotor is rotated in the correct direction. 3. 星型卸料阀 a. 检查转子的旋转方向是否正确。 4. Submerged pump(4306-PSU-001) a. Check whether there is debris in the lower pump tank and the pipeline blockage, Whether the outlet butterfly valve (65BF) is normal; b. Check whether the fastening screws of the equipment and motor are loose, whether the bearing is intact, and whether the lubricating oil is intact; c. Check whether the electrical part of the equipment is in good condition and whether the motor shell is well grounded; d. Check whether the steering of the motor is correct; e. Check whether the pump head is under the ground trough liquid level. 4. 液下泵(4306-PSU-001) a. 检查液下泵槽内有无杂物，管路堵塞情况，出口蝶阀(65BF)是否正常； b. 检查设备、电机的紧固螺丝是否松动，轴承是否完好，润滑油是否完好； c. 检查设备电气部分是否完好，电机外壳是否接地良好； d. 检查电机的转向是否正确； e. 检查泵头是否处于地槽液位下。 5. Tape (4303-BCV-006) (4303-CHE-008/009/010), spiral conveyor (4303-CNV-001) (4303-CHE-014), filter press (4303-FIL-001) (4303-CHE-015) a. Check whether the feeding chute is in good condition; b. Check whether the blocking detection switch of the feeding chute is normal. 5. 胶带机(4303-BCV-006)下料溜槽(4303-CHE-008/009/010)、螺旋输送机(4303-CNV-001)卸料溜槽

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		(4303-CHE-014)、压滤机(4303-FIL-001)下料溜槽(4303-CHE-015) a. 检查各下料溜槽是否完好； b. 检查下料溜槽的堵料检测开关是否正常。 6. Import and outlet valve of fan (4301-FAN-005/201) a. Check whether the inlet and outlet valve (****,****) is normal. 6. 风机 (4301-FAN-005/201)进出口阀门 a. 检查进出阀门(****,****)是否正常。 7. Other a. Circulation water inlet gate valve (32GA), produce new water stop valve (25GB) butterfly valve (LV011, LV012), circulating pump (4303-PSO-001/002/003/004) inlet and outlet butterfly valve (350BF, 300BF), filter press (4303-FIL-001) (100PN) and check valve (100CN), filter pump (4303-PSL-001/002) inlet clamp valve (100PN, 80PN), check inlet and outlet pipeline, valve installation meet the requirements; b. Check whether the liquid level control components of primary empty tower washtower (4303-TOW-001-1), secondary packing washtower (4303-TOW-001-2) and efficient multi-tube dust collector (4303-DUC-001) the level meter is normal. 7. 其它 a. 循环水入口闸阀门(32GA)，生产新水截止阀(25GB)蝶阀(LV011,LV012)，循环泵(4303-PSO-001/002/003/004)出入口蝶阀(350BF,300BF)，压滤机(4303-FIL-001)入口夹管阀(100PN)和止回阀(100CN)，压滤泵(4303-PSL-001/002)出入口夹管阀(100PN,80PN)，检查出入口管线及附属管线，阀门安装是否符合要求； b. 检查一级空塔洗涤塔 (4303-TOW-001-1)、二级填料洗涤塔 (4303-TOW-001-2)、高效多管除尘器(4303-DUC-001)料位计是否正常。
14	Inspect Materials Level 检查物料料位	1. Check whether the material is stored on the tape machine (4303-BCV-006), and there is no adhesive deposit on the surface of the head and tail wheel and the belt joint; 2. Check the multi-efficiency dust collector (4303-DUC-001) for dust; 3. Check whether the liquid level of the primary empty tower washing tower (4303-TOW-001-1) and the secondary filler

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		washing tower (4303-TOW-001-2) meets the requirements; 4. Check whether the material is stored in the filter press (4303-FIL-001), Whether the hydraulic oil is sufficient, Whether the feeding chute (4303-CHE-015) has a filter cake; 5. Check whether the material is stored in the screw conveyor (4303-CNV-001); 6. Check the cloth bag of the cloth bag dust collector (4301-DUC-201) and the ash in the horizontal flue in front of the cloth bag dust collector. 7. Check whether the material is stored in the pipe chain conveyor(4303-PCC-001) 1. 检查胶带机(4303-BCV-006)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在; 2. 检查高效多管除尘器(4303-DUC-001)是否存尘; 3. 检查一级空塔洗涤塔 (4303-TOW-001-1)，二级填料洗涤塔 (4303-TOW-001-2)液位是否满足要求; 4. 检查压滤机(4303-FIL-001)内是否存料，液压油是否足够，下料溜槽(4303-CHE-015)是否存有滤饼; 5. 检查螺旋输送机(4303-CNV-001)内是否存料; 6. 检查布袋除尘器 (4301-DUC-201)的布袋沾粘物料，布袋除尘器前水平烟道中的积灰; 7. 检查管链输送机(4303-PCC-001)内是否存料。
15	Start-Up Of The Grate Bed Post 下篦床岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, Each equipment needs to be switched on later; At the same time, Confirm that the process behind the lower grate bed post has been started and has the feeding conditions: Open the secondary packing washer tower (4303-TOW-001-2) → Open the water stop valve and butterfly valve (LV012,25GB), load the tank into 3/4 full clear water, then close the butterfly valve (LV012), open the circulating pump for 3 minutes, then release the clear water from the outlet butterfly valve (50BF, 300BF) of the waste water in the washing tower; close the outlet butterfly valve (50BF, 300BF), open the filling water butterfly valve (LV012), and inject the fresh water again in the circulating tank until full → Close the water supply butterfly valve (LV012), open the water filling valve to adjust the height of the floating ball switch, and do not adjust arbitrarily after adjustment → Open the circulation pump (4303-PSO-003/004) outlet butterfly valve (350BF) → Open the circulation pump (4303-PSO-003/004) → Open the primary tower (4303-TOW-001-1) → Open the water stop valve and butterfly valve (LV011,25GB), load the

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		tank into 3/4 full of water, then close the butterfly valve (LV011), open the circulating pump for 3 minutes, then release the water from the butterfly valve (50BF, 300BF) of the water in the washing tower; close the waste water outlet butterfly valve (50BF, 300BF), open the water butterfly valve (LV011), and inject the water again in the circulating tank until full → Close the water supply butterfly valve (LV011), open the water filling valve to adjust the height of the floating ball switch, and do not adjust arbitrarily after adjustment → Open the circulation pump (4303-PSO-001/002) outlet butterfly valve (350BF) → Open the circulation pump (4303-PSO-001/002) → Open the cooling water of the induced draft fan (4303-FAN-005), open the outlet clamp valve (100PN), open the induced draft fan, and open the inlet clamp valve of the induced draft fan (80PN) → Open the high efficiency multi-tube dust collector (4303-DUC-001), first turn off the outlet clamp valve (100PN) of the small induced draft fan (4303-FAN-005), gradually open the large clamp valve (100PN) after checking no abnormal phenomenon, and release the ash on time, generally no more than 4 hours / time → Open the adhesive tape machine (4303-BCV-001) → Open the adhesive tape machine (4303-BCV-005) → Open the pipe chain conveyor (4303-PCC-001) → Open the fan (4303-FAN-201), when the fan operates normally without load, open the inlet valve of the fan (4301-FAN-201) (****) → Open the bag filter (4303-DUC-201) → Open the star-type discharge valve of the cloth bag dust collector → Start the spiral conveyor in the cloth bag dust collector → Adjust the spray pressure of the bag filter, the setting point, pulse frequency and pulse width (after the first startup setting, the next startup will not adjust), start the bag filter → Open the inlet valve of the front end of the dust collector (****), and enter the dust and smoke gas → Observe the rising resistance of the dust collector and the discharge of the exhaust outlet. When the new filter bag is put into use, there will be trace dust escape from the discharge outlet → When the resistance of the bag filter rises to 1000Pa~1200Pa, start the ash removal controller for ash removal, and observe the pressure drop → Open the vibrator (4303-RAP-001) → Open the spiral conveyor (4303-CNV-001) star valve → Open the screw conveyor (4303-CNV-001) → Open the filter press (4303-FIL-001) and open the pressure filter pressure pump (4303-PSL-001 / 002) → Press the Start-Up Push-up button, Oil pump starts working, Then press the "press" button, The piston pushes the pressing plate is pressed, When the pressing force reaches the adjusted high point pressure, The hydraulic system automatically stops and maintains pressure, Open the filter

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		press inlet clamp valve (100PN) and check valve (100CN) → Slowly open the pressure filter pump inlet clamp valve (100PN, 80PN), Each slurry through the feed hole on the thrust plate, Under pressure filtration at a feed pressure no greater than 0.6MPa, Gradually form the filter cake → After the filter cake is filled, shut down the filter pump (4303-PSL-001/002) and the inlet clamp valve (100PN) and the check valve (100CN) → Hold down the "release" button on the operation panel, the press plate will automatically stop after returning in place, and the pull hook moves forward to unload the reciprocating plate → Open the double-layer ash discharge valve (4303-ADG-001~028) → Open the adhesive tape machine (4303-BCV-006). 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；**同时，确认下篦床岗位后面的工序已经开车并具备给料条件:** 开启二级填料洗涤塔(4303-TOW-001-2)→打开补水的截止阀和蝶阀(LV012,25GB)，将水箱装入3/4满清水，然后关闭蝶阀(LV012)，开启循环水泵3分钟，然后将废水排放的出口蝶阀(50BF,300BF)释放洗涤塔中之清水；关闭废水排放的出口蝶阀(50BF,300BF)，打开补水蝶阀(LV012)，在循环水槽内再次注入清水直到满为止→关闭补水蝶阀(LV012)，打开补水阀调整浮球开关高度，调整好后不得任意再调整→打开循环泵(4303-PSO-003/004)出口蝶阀(350BF)→开启循环泵(4303-PSO-003/004)→开启**一级空塔洗涤塔**(4303-TOW-001-1)→打开补水的截止阀和蝶阀(LV011,25GB)，将水箱装入3/4满清水，然后关闭蝶阀(LV011)，开启循环水泵3分钟，然后将废水排放的出口蝶阀(50BF,300BF)释放洗涤塔中之清水；关闭废水排放的出口蝶阀(50BF,300BF)，打开补水蝶阀(LV011)，在循环水槽内再次注入清水直到满为止→关闭补水蝶阀(LV011)，打开补水阀调整浮球开关高度，调整好后不得任意再调整→打开循环泵(4303-PSO-001/002)出口蝶阀(350BF)→开启循环泵(4303-PSO-001/002)→**打开**引风机(4303-FAN-005)**冷却水，打开出口夹管阀**(100PN)，开启**引风机，打开引风机入口夹管阀**(80PN)→开启高效多管除尘器(4303-DUC-001)，先关小引风机(4303-FAN-005)出口夹管阀(100PN)，检查无异常现象后逐渐开大夹管阀(100PN)，按时出灰，一般不超过4小时/次→开启胶带机(4303-BCV-001)→开启胶带机(4303-BCV-005)→开启管链输送机(4303-PCC-001)→开启风机 (4303-FAN-201)，待风机无负荷运转正常，打开风机 (4301-FAN-201)入口阀门

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		(****)→开启布袋除尘器 (4303-DUC-201)→开启布袋除尘器的星型卸料阀→启动布袋除尘器内的螺旋输送机→调整布袋除尘器喷吹压力、开始喷吹的设置点、脉冲频率和脉冲宽度(首次开机设定后，下次开机不再调整)，启动布袋除尘器→打开除尘器前端进口阀门(****)，通入含尘烟气→观察除尘器运行阻力上升情况及排气口的排放情况，在新滤袋投入使用时，排放口将会有微量粉尘逸出→当布袋除尘器阻力上升到1000Pa ~ 1200Pa时，启动清灰控制器进行清灰，并观察压力下降情况→开启振打器(4303-RAP-001)→开启螺旋输送机(4303-CNV-001)星型阀→开启螺旋输送机(4303-CNV-001)→开启压滤机(4303-FIL-001)→开启压滤泵(4303-PSL-001/002)，按下“启动”按钮，油泵开始工作，然后再按一下“压紧”按钮，活塞推动压紧板压紧，当压紧力到达调定高点压力后，液压系统自动跳停并保持压力，打开压滤机入口夹管阀(100PN)和止回阀(100CN)→缓慢打开压滤泵出入口夹管阀(100PN,80PN)，料浆通过止推板上的进料孔进去各个滤室，在进料压力不大于0.6MPa下加压过滤，逐渐形成滤饼→压滤机滤饼充满后，关停压滤泵(4303-PSL-001/002)和入口夹管阀(100PN)和止回阀(100CN)→按住操作面板上的“松开”按钮，压板退回到位后会自动停止，同时拉板钩前进进行往复拉板卸料→开启双层卸灰阀(4303-ADG-001 ~ 028)→开启胶带机(4303-BCV-006)。
16	The Grate Bed Post Stops 下篦床岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order and material discharge order, Each equipment needs to stop: Stop tape machine (4303-BCV-006)→ After grate (4303-CHG-001), Close the double-layer ash discharge valve (4303-ADG-001~028)→ Close the inlet clamp valve of the induced draft fan (80PN), stop the induced draft fan, and close the outlet clamp valve of the induced draft fan (4303-FAN-005) (100PN)→ Stop circulation pump (4303-PSO-001/002)→ Stop the primary air tower washer (4303-TOW-001-1), close the circulation pump (4303-PSO-001/002) outlet butterfly valve (350BF)→Turn off the circulation pump (4303-PSO-003/004)→ Stop the secondary packing washing tower (4303-TOW-001-2), close the circulation pump (4303-PSO-003/004) outlet butterfly valve (350BF)→The primary empty tower tower (4303-TOW-001-1) and the secondary filler tower (4303-TOW-001-2) have no bottom precipitation, Close the pressure filter pressure pump (4303-PSL-001/002) and stop the filter press (4303-FIL-001)

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№	Stage	Key Points
		⟹Close the inlet clamp valve (80PN), stop the filter pump (4303-PSL-001/002), the outlet clamp valve (100PN) and close the filter press (4303-FIL-001) (100CN)⟹ Stop the efficient multi-tube dust collector (4303-DUC-001) ⟹Stop spiral conveyor (4303-CNV-001) star valve ⟹ Stop spiral conveyor (4303-CNV-001)⟹ Stop the bag dust collector before the bag filter (4303-DUC-201)⌊Close the front inlet valve of the dust collector (****) and stop the dust and smoke gas⟹ After the collected dust and dust, Stop the spiral conveyor in the cloth bag dust collector⟹ Stop the star-type discharge valve of the cloth bag dust collector⌋⟹ Close the inlet valve of the fan (4303-FAN-201) (****) and stop the fan (4303-FAN-201)⟹ Stop pipe chain conveyor (4303-PCC-001)⟹ Stop the tape machine (4303-BCV-005)⟹ Stop the tape machine (4303-BCV-001)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序和物料排净顺序停车，各设备都需要延时停止： 停止胶带机(4303-BCV-006)⟹链篦机(4303-CHG-001)无落灰后，关闭双层卸灰阀(4303-ADG-001～028)⟹关闭引风机入口夹管阀(80PN)，停止引风机，关闭引风机(4303-FAN-005)出口夹管阀(100PN)⟹停止循环泵(4303-PSO-001/002)⟹停止一级空塔洗涤塔(4303-TOW-001-1)，关闭循环泵(4303-PSO-001/002)出口蝶阀(350BF)⟹关闭循环泵(4303-PSO-003/004)⟹停止二级填料洗涤塔(4303-TOW-001-2)，关闭循环泵(4303-PSO-003/004)出口蝶阀(350BF)⟹一级空塔洗涤塔(4303-TOW-001-1)和二级填料洗涤塔(4303-TOW-001-2)无塔底沉淀后关闭压滤泵(4303-PSL-001/002)和停止压滤机(4303-FIL-001)⟹关闭压滤泵入口夹管阀(80PN)，关停压滤泵(4303-PSL-001/002)，关闭压滤泵出口夹管阀(100PN)；关闭停止压滤机(4303-FIL-001)入口夹管阀(100PN)和止回阀(100CN)，关闭电源⟹停止高效多管除尘器(4303-DUC-001)⟹停止螺旋输送机(4303-CNV-001)星型阀⟹收集的烟尘清灰结束且输送完成后，停止螺旋输送机(4303-CNV-001)⟹布袋除尘

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№	Stage	Key Points
		器前端无含尘烟气后，停止布袋除尘器 (4303-DUC-201) 关闭除尘器前端进口阀门(****)，停止含尘烟气 ⟹ 收集的烟尘清灰结束且输送完成后，停止布袋除尘器内的螺旋输送机 ⟹ 停止布袋除尘器的星型卸料阀 ⟹ 关闭风机 (4303-FAN-201)入口阀门(****)，停止风机 (4303-FAN-201) ⟹ 停止管链输送机(4303-PCC-001) ⟹ 停止胶带机(4303-BCV-005) ⟹ 停止胶带机(4303-BCV-001) ⟹ 打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, tape (4303-BCV-006), double-layer ash discharge valve (4303-ADG-001-028), induced draft fan (4303-FAN-005), primary tower washer (4303-TOW-001-1), secondary filler washer (4303-TOW-001-2), circulating pump (4303-PSO-001/002/003 (4303-PSL-001/004), filter press (4303-FIL-001), efficient filter (4303-DUC -001), star discharge valve, spiral conveyor (4303-CNV-001), adhesive tape machine (4303-BCV-001/005), cloth bag filter (4303-DUC-201), pipe chain conveyor (4303-PCC-001), fan (4303-FAN-201), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close the inlet and outlet pipe clamp valve for the induced draft fan (4303-FAN-005) (100PN, 80PN), circulating pump (4303-PSO-001/002/003/004) outlet butterfly valve (350BF), pressure filter pump (4303-PSL-001/002) inlet clamp valve (100PN, 80PN), filter press (4303-FIL-001) inlet clamp valve (100PN) and check valve (100CN), bag filter (4303-DUC-201) front end inlet valve (****), fan (4303-FAN-201) inlet and out valve (****, ****), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, and the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭胶带机(4303-BCV-006)、双层卸灰阀(4303-ADG-001 ~ 028)、引风机(4303-FAN-005)、一级空塔洗涤塔(4303-TOW-001-1)、二级填料洗涤塔(4303-TOW-001-2)、循环泵(4303-PSO-001/002/003/004)、压滤泵(4303-PSL-001/002)、压滤

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№	Stage	Key Points
		机(4303-FIL-001)、高效多管除尘器(4303-DUC-001)、星型卸料阀、螺旋输送机(4303-CNV-001)、胶带机(4303-BCV-001/005)、布袋除尘器 (4303-DUC-201)、管链输送机(4303-PCC-001)、风机 (4303-FAN-201)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭引风机(4303-FAN-005)出入口夹管阀(100PN,80PN)、循环泵(4303-PSO-001/002/003/004)出口蝶阀(350BF)、压滤泵(4303-PSL-001/002)出入口夹管阀(100PN,80PN)、压滤机(4303-FIL-001)入口夹管阀(100PN)和止回阀(100CN)、布袋除尘器 (4303-DUC-201)前端进口阀门(****)、风机 (4303-FAN-201)进出口阀门(****,****)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
17	Permits &Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. The Grate Bed Post 下篦床岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

The grate bed posts in the chloride plant
CP 下篦床岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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